

Lab 4: VibeCoding — Meteor Dodge Report & Chat History

- **Course / Assignment:** Lab 4: VibeCoding
- **Student GitHub:** [Pradhyut21](https://github.com/Pradhyut21)
- **Forked Repository:** <https://github.com/Pradhyut21/38-meteor-dodge>
- **Tool Used:** Google Antigravity (Gemini 3.8 Flash Pair Programmer)

1. Summary of Deliverables

Deliverable	Location / Filename	Description
Video: Before	Lab-4/before.mp4 / before.mp4	10.0 seconds (300 frames, 30 FPS) showing gameplay before fixes, highlighting the bug where pressing SPACE during flight fails to fire defensive lasers.
Video: After	Lab-4/after.mp4 / after.mp4	10.0 seconds (300 frames, 30 FPS) showing gameplay with all 4 tasks implemented: laser firing, meteor splitting, shield pickup/absorption, and dynamic multiplier scaling.
Updated Code	game/ package	All bug fixes and features implemented and tested.
Git Commits	Repository Commit History	Clean, individual commits corresponding to each task.

2. Tasks & Bug Resolution Walkthrough

Task 1: Fix the laser firing / input state bug

- **Identified Bug:** In `game/game_engine.py`, the `handle_events` method checked `if event.key == pygame.K_SPACE: if self.game_over: self.reset() else: self.started = True`. When flight was already active (`self.started == True`), pressing space did nothing instead of firing a defensive laser projectile.
- **Solution:**
 1. Created `game/laser.py` with `Laser` class containing position, vertical velocity (`vy = -10`), collision check (`collides`), and glowing beam rendering.
 2. Added `fire()` method in `game/ship.py`.
 3. Updated `game/game_engine.py` to spawn laser projectiles on spacebar input during active flight, update projectile motion, and detect collisions with meteors.
- **Git Commit:** Task 1: Fix laser firing and spacebar input state bug (e71960f)

Task 2: Implement Meteor Splitting on Impact

- **Requirement:** Destroying large meteors causes them to fracture into smaller child fragments diverging outwards, while small meteors dissolve completely.
- **Solution:**
 1. Updated `game/meteor.py` with `split()` method.

2. Meteors with **radius > 16** fracture into two smaller fragments (**radius * 0.58**) with diverging horizontal velocities (**vx ± random.uniform(1.8, 3.2)**).
3. Small meteors (**radius <= 16**) return empty lists and dissolve.
4. Updated laser collision logic in **game/game_engine.py** to remove parent meteor and append child fragments.
 - **Git Commit: Task 2: Implement meteor splitting on laser impact (fdcd4bb)**

Task 3: Implement Collectible Shield Power-Up Orbs

- **Requirement:** Ship is destroyed on any contact. Introduce drifting energy orbs that the player can collect to gain a temporary shield barrier absorbing one collision.
 - **Solution:**
1. Created **game/shield_orb.py** with drifting sinusoidal wobble motion, pulsing outer ring, and collision detection.
 2. Added **has_shield** attribute and animated translucent glowing shield barrier in **game/ship.py**.
 3. In **game/game_engine.py**, periodically spawn shield orbs. When collected by the ship, activate **ship.has_shield**.
 4. When a meteor hits the ship while shield is active, absorb the hit, remove the meteor, and consume the shield without triggering game over.
 - **Git Commit: Task 3: Implement collectible shield power-up orbs (fbf6027)**

Task 4: Implement Consecutive Survival Multipliers

- **Requirement:** Introduce escalating multiplier boosting score gain for every 10 seconds of continuous survival without taking a hit.
 - **Solution:**
1. In **game/game_engine.py**, tracked **survival_frames** and continuous **streak_frames**.
 2. Multiplier calculated as **1 + (streak_frames // 600)** (every 600 frames at 60 FPS is 10 seconds).
 3. Score increments each frame by the active multiplier (**self.score += self.multiplier**).
 4. If a meteor is absorbed by the shield, streak resets to 0 and multiplier resets to 1x.
 5. Updated HUD in **draw()** to display elapsed time, total score, and active multiplier (**x1, x2**, etc.) in gold/accent colors.
 - **Git Commit: Task 4: Implement consecutive survival score multipliers (04f4283)**

3. Commit History

```
e34a4cc Docs: Update folder structure in README.md
04f4283 Task 4: Implement consecutive survival score multipliers
fbf6027 Task 3: Implement collectible shield power-up orbs
fdcd4bb Task 2: Implement meteor splitting on laser impact
e71960f Task 1: Fix laser firing and spacebar input state bug
b168286 Update README.md
49607ea Add files via upload
```

4. Video Recording Verification

Both videos were generated using headless Pygame frame capture with OpenCV VideoWriter (**mp4v** codec) at 30 FPS:

- **before.mp4**: 300 frames, 10.0s duration, 700x520 resolution. Demonstrates ship flight, falling meteors, and spacebar press failing to fire lasers.
- **after.mp4**: 300 frames, 10.0s duration, 700x520 resolution. Demonstrates laser fire, meteor splitting into fragments, shield orb collection, shield absorption, and score multiplier HUD scaling.